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A Supplementary Information

Note S1.1 Metrics on full test set and for 10k materials predicted most stable

Unlike table 1 which evaluates model performance on the subset of unique WBM protostructures, table S1 includes the full WBM test set of 257k materials. The 44k additional materials in table S1 excluded in table 1 comprise 11 175 discarded due to having a matching protostructure in MP plus another 32 784 materials which are protostructure duplicates of another WBM material with lower energy. The most noteworthy difference between the two tables is a drop in DAF for all models in table S1 compared to table 1. MACE for example achieves a DAF of 3.5 on the full test set (table S1) compared to 3.85 on the subset of 215.5k materials with unique protostructures (table 1). This $\sim 10\%$ increase is largely due to a $\sim 10\%$ decrease in the fraction of materials below the MP convex hull: 15.3% (32,942 out of 215,488) in table 1 vs 16.7% (42,825 out of 256,963) in the full dataset (table S1). Since DAF is the ratio of the model’s precision for stability prediction to the prevalence of stable structures, a lower prevalence results in a higher DAF.

While protostructures are non-trivial to match, thus potentially introducing bias into the test set by trying to deduplicate them, we still opted to feature table 1 in the main text since the removal of overlapping protostructures with MP makes it more closely reflect a model’s true ability to extrapolate to out-of-domain materials. Protostructure assignment is based on a combination of Aflow-style prototype labels and the chemical system [81] we use the `get_protostructure_label_from_spglib` implemented in `aviary` [82]. This string encodes the crystal’s prototype and chemical system and is invariant to symmetry preserving changes to the structure. Consequently structures with different lattice parameters are flagged as protostructure duplicates as they would both be expected to relax to the same ground state in experimental conditions.

A real-world discovery campaign is unlikely to validate all stable predictions from a given model as we did in table 1. Presumably, it will rank model predictions from most to least stable and follow that list as far as time and compute budget permits. Assuming that increases in compute resources will allow average future discovery campaigns to grow in scope, we believe 10k model validations to be a reasonable scope for average campaigns. This is what table S2 simulates by calculating classification and regression metrics for the 10k test set materials predicted to be most stable by each model. We again show dummy performance in the bottom row. Note that each model is now evaluated on a different slice of the data. However, the bottom row still shows dummy performance across the whole dataset.

Note S1.2 ROC Curves

A material is classified as stable if the predicted $E_{\text{above hull}}$ is below the stability threshold. Since all models predict E_{form} , they are insensitive to changes in the threshold t . The receiver operating characteristic (ROC) curve for each model is plotted in fig. S1. The diagonal ‘No skill’ line shows the performance of a dummy model that randomly ranks material stability.

Note S1.3 Parity Plots

Figure S2 shows that all models do well for materials far below the convex hull (left side of the plot). Performance for materials far above the convex hull is more varied with occasional underpredictions of the energy of materials far above the convex hull (right side). All models suffer most in the mode of the distribution at $x = 0$.

Two models stand out as anomalous to the general trends.

Wrenformer is the only model with a large number of severe energy overpredictions at $x = 0$ along the positive y axis. We investigated these failure cases in more detail and found these overpredictions to be dominated by spacegroup 71 with poor representation in the training data. Digging into the MP-crystals-2022.10.28 dataset we see that on post analysis there are a large number of A_2BC compounds in spacegroup 71 in the v2022.10.28 that are trapped in local minima and thus much higher in energy. Given that Wrenformer nominally predicts the energy the global minima conditioned on the protostructure these are unsuitable for training Wrenformer and their inclusion leads to the errors seen on similar, correctly relaxed structures, in the WBM test set.

The other anomalous model is MACE with several severe underpredictions at $x = 0$ along the negative y axis. We investigated these points for common traits in composition or crystal symmetry but noticed no pattern.

Beyond these MACE outliers visible in the plot, MACE exhibits another rare but reproducible type of failure case, in which the final predicted energy after relaxation is off by several orders of magnitude. The largest ‘derailed’ prediction was -10^{22} eV/atom for `wbm-3-31970` (formula H_2Ir). In each case, the MACE relaxation exhausted the maximum number of ionic steps set to 500 and caused a volume implosion from initial cell volumes of hundreds to relaxed cell volumes of tens of cubic Angstrom. Using the checkpoint trained on the M3GNet dataset which we received from the MACE authors, this failure mode occurred for several hundred of the 250k test set crystals. Using the checkpoint we trained ourselves on the MPtrj dataset, it affects only 44 test crystals, suggesting that these holes in the MACE PES can perhaps be fully plugged by further increasing the training set or even changing the loss function. Further analysis is ongoing. Since these derailed values are easily identified in practice when actually performing a prospective discovery campaign, we excluded

Model	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	TPR $\uparrow$	TNR $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R ² $\uparrow$
eqV2 S DeNS	0.798	4.362	0.748	0.927	0.855	0.941	0.035	0.084	0.785
ORB MPtrj	0.755	4.188	0.719	0.911	0.795	0.936	0.043	0.090	0.752
SevenNet	0.719	3.804	0.653	0.893	0.800	0.912	0.046	0.090	0.750
MACE	0.668	3.400	0.583	0.867	0.781	0.885	0.055	0.099	0.698
CHGNet	0.612	3.038	0.521	0.839	0.740	0.859	0.061	0.100	0.690
M3GNet	0.576	2.647	0.454	0.802	0.788	0.804	0.072	0.115	0.588
ALIGNN	0.565	2.921	0.501	0.829	0.649	0.866	0.092	0.154	0.274
MEGNet	0.513	2.699	0.463	0.813	0.574	0.862	0.128	0.204	-0.277
CGCNN+P	0.510	2.398	0.411	0.779	0.670	0.801	0.108	0.178	0.027
CGCNN	0.510	2.631	0.451	0.807	0.587	0.852	0.135	0.229	-0.624
Wrenformer	0.479	2.130	0.365	0.741	0.693	0.751	0.105	0.182	-0.020
BOWSR	0.437	1.836	0.315	0.702	0.711	0.680	0.114	0.164	0.142
Voronoi RF	0.344	1.509	0.259	0.665	0.511	0.697	0.141	0.206	-0.316
Dummy	0.194	1.000	0.168	0.680	0.231	0.770	0.120	0.181	0.000

Table S1: Same as table 1 but computed for all 257k structures in the WBM test set, no duplicate protostructures excluded. The prevalence of stable structures is 16.7% (42,825 out of 256,963), placing a ceiling of 6 on the maximum possible DAF.

Model	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R ² $\uparrow$
eqV2 S DeNS	0.983	6.326	0.967	0.967	0.031	0.091	0.823
ORB MPtrj	0.971	6.173	0.944	0.944	0.037	0.098	0.801
SevenNet	0.945	5.857	0.895	0.895	0.054	0.124	0.700
CHGNet	0.920	5.567	0.851	0.851	0.063	0.095	0.816
MACE	0.888	5.221	0.798	0.798	0.087	0.165	0.508
M3GNet	0.868	5.020	0.767	0.767	0.101	0.158	0.551
ALIGNN	0.748	3.905	0.597	0.597	0.176	0.247	0.081
CGCNN	0.745	3.880	0.593	0.593	0.165	0.230	0.181
CGCNN+P	0.736	3.813	0.583	0.583	0.200	0.275	-0.076
Wrenformer	0.721	3.691	0.564	0.564	0.182	0.239	0.138
BOWSR	0.664	3.252	0.497	0.497	0.259	0.320	-1.172
MEGNet	0.632	3.022	0.462	0.462	0.304	0.336	-0.908
Voronoi RF	0.551	2.487	0.380	0.380	0.349	0.417	-1.012
Dummy	0.194	1.000	0.168	0.680	0.120	0.181	0.000

Table S2: Stability prediction metrics for the 10k materials predicted to be most stable by each model. Every model picks out a different slice of the test set as the 10k materials it believes to be furthest below the known convex hull.

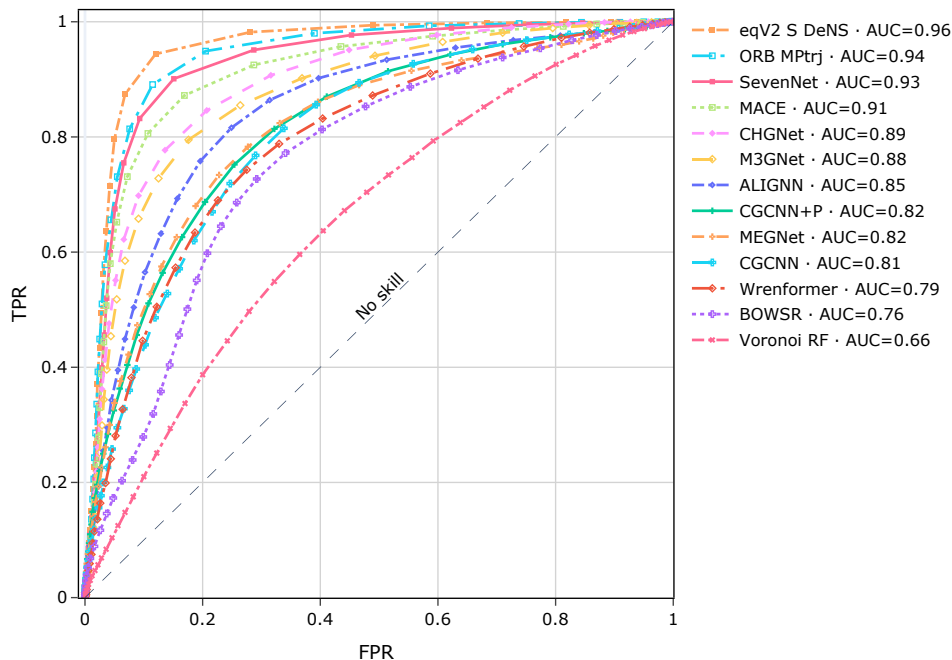


Figure S1: Receiver operating characteristic (ROC) curve for each model. The false positive rate (FPR) on the x axis is the fraction of unstable structures classified as stable. The true positive rate (TPR) on the y axis is the fraction of stable structures classified as stable.

them from the MACE parity plot and all other downstream analyses.

Note S1.4 Hull Distance Box plot

Figure S4 shows a box-plot of the errors in predicting the distance to the convex hull. The box demarks the interquartile range and the whiskers show the 5th and 95th percentiles. BOWSR has the largest median error, while Voronoi RF has the largest IQR. Looking at the models we see that there is a systematic trend to underpredict the distance to the convex hull. This trend can be rationalized given the bias in the Materials Project towards stable and therefore low formation energy structures. In contrast, due to its construction the WBM test set is distributionally distinct with a higher average formation energy. Therefore we propose that models trained on the Materials Project are likely result in such systematic underpredictions and this affect should be able to be addressed by training on additional data from a similar distribution to WBM i.e. Alexandria [7].

Note S1.5 Classification Histograms using Model-Predicted Energies

Figure S5 shows histograms of the model predicted distances to the convex hull colored by the stability classification. These plots allow for practitioners to assess how the accuracy varies for materials depending on how far above or below the hull they are predicted to lie. The

most pronounced affect we see from the histograms is that the CGCNN+P histogram is more strongly peaked than CGCNN’s giving much better agreement with the actual DFT ground truth distribution of hull distances for the test set. This explains why CGCNN+P performs better as a regressor, but also reveals how it can perform simultaneously worse as a classifier. By moving predictions closer to the stability threshold at 0 eV/atom above the hull, even small errors are significant enough to tip a classification over the threshold.

Note S1.6 Measuring extrapolation performance from WBM batch robustness

As a reminder, the WBM test set was generated in 5 successive batches, each step applying another element replacement to an MP source structure or a new stable crystal generated in one of the previous replacement rounds. The likelihood by which one element replaces another is governed by ISCD-mined chemical similarity scores for each pair of elements. Naively, one would expect model performance to degrade with increasing batch count, as repeated substitutions should on average ‘diffuse’ deeper into uncharted regions of material space, requiring the model to extrapolate more. We observe this effect for some models much more than others.

Figure S6 shows the rolling MAE as a function of distance to the convex hull for each of the 5 WBM rounds of elemental substitution. These plots show a stronger per-